

Adaptive Modulation and Coding (AMC) in 5G NR

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Abstract—In many 5G New Radio (NR) systems, variables such as coding rate and modulation order are constantly modified to fit the channel quality, this approach is called Adaptive Modulation and Coding (AMC). This study utilizes MATLAB to evaluate the AMC performance in a 5G link simulation. We use Exponential Effective SINR Mapping (EESM) to compress the Signal to Noise Ratio (SNR) to produce a single effective SNR, this variable then determines the Channel Quality Indicator (CQI), which directly corresponds to a Modulation and Coding Scheme (MCS) in accordance with 3GPP TS 38.214. We evaluate AMC under AWGN and Rayleigh, and we compare both to a fixed 64-QAM (MCS 17) baseline under the same graph conditions.

I. INTRODUCTION

The development of fifth-generation (5G) New Radio (NR) has brought ultra-high speeds, low latency, and massive connectivity [1]. The various requirements are effectively met by 5G technology, and one prominent link adaptation technique used by 5G technology is *Adaptive Modulation and Coding* or AMC.

When the channel conditions are good, the high-order constellations, 64-QAM or 256-QAM, are employed to optimize the bandwidth utilization, while when the channel condition is poor, lower-order constellations, QPSK with greater coding rates, are used to ensure link reliability.

This project involves simulation and analysis of AMC in 5G NR using MATLAB. The aim is to determine the values of the throughput, spectral efficiency, and BLER as a function of SNR and distance for both AWGN and Rayleigh channels. As part of 3GPP standard requirements, the adaptive scheme will make use of predetermined MCS values as specified in TS 38.214 [2] to allow an auto-switching mechanism between the modulation order and coding rate as a function of the measured SNR.

II. BACKGROUND

In 5G NR, the AMC relies on feedback between the User Equipment (UE) and the Base Station (gNodeB). The CQI will be measured by the UE for the downlink channel, and this will be reported to the base station. The base station will then make an MCS decision, which will be an index, out of a standard list mentioned as 3GPP TS 38.214. The standard list will define an appropriate index, which will relate to a given Modulation scheme, like QPSK, 16-QAM, 64-QAM, or 256-QAM, as well as a coding rate, which will define the efficient bandwidth utilization. In our MATLAB implementation, these standard MCS mappings are directly

used to drive AMC selection, ensuring that the modulation and coding rate adapt in accordance with 5G-compliant link adaptation procedures.

A. Adaptive Modulation and Coding (AMC)

Adaptive Modulation and Coding (AMC) is an important link adaptation scheme practiced by wireless communication networks. The aim is to maximize the data rate, maintaining a desired, or target, Block Error Rate (BLER), by assigning the optimum value of the modulation order and coding rate, according to the channel condition. In 5G NR, the channel quality measured is the SNR, which takes into consideration fading.

As SNR values change over each subcarrier in an OFDM signal, the receiver sends a Channel Quality Indicator (CQI) message to the Base Stations (gNodeB) based on an Effective SNR. Within this thesis, EESM is employed to reduce a set of SNR values, associated with each subcarrier, to a representative Effective SNR suitable for CQI feedback. The Base Stations, later, map this CQI to a Modulation and Coding Scheme (MCS) index identified by the 3GPP TS 38.214 standardized MCS values. Every selected MCS index represents a pair of values, including a type representing QPSK, 16-QAM, 64-QAM, or 256-QAM, as well as a specified coding rate.

The objective of AMC is to enhance spectral efficiency under strong channels by applying high constellation modulations with less coding, as well as ensure reliability under weak channels by applying low constellation modulations with stronger coding. Such an adaptive scheme enables reliable communication even in differing channel conditions and interference levels.

B. Fixed Modulation and Coding (FMC)

Fixed Modulation and Coding, FMC, is a 5G NR communication mode whereby a system communicates with a fixed coding rate and a fixed coding modulation regardless of channel conditions. Such a system, rather than relying on an Adaptation Modulation and Coding, AMC, system, adopts a fixed coding mode whereby, rather than adapting to varying signal quality, a fixed coding scheme, such as QPSK coding, 0.5 code rate, or 16-QAM, 0.75 code rate, among others, is used to enhance signal quality as opposed to changing coding rates. FMC, rather than providing an optimal solution to communication, as solving full channel capacity, offers a less-than-optimal solution as it doesn't maximize channel

capacity, meaning that whether the channel is good or bad, the communication link doesn't maximize channel capacity. A recent study by Méndez-Monsanto et al., 2024 [3], highlights the detailed results relating to BLER-SNR curves, as detailed by various 5G NR MCS, as a reference solution when relating to a communication link adopting a fixed-MCS, as opposed to an adaptative scenario, AMC, communication solution.

C. 5G NR Physical Layer and Modulation Schemes

The physical layer of 5G NR relies on Orthogonal Frequency-Division Multiplexing (OFDM) as its waveform [4], with scalable subcarrier spacing and flexibility of resource allocation. The modulation schemes [5] available include:

- QPSK (Quadrature Phase Shift Keying) – balanced reliability and efficiency.
- 16-QAM, 64-QAM, 256-QAM – higher-order constellations offering higher data rates but increased susceptibility to noise.

The coding scheme in 5G NR involves the use of Low-Density Parity-Check or LDPC codes, which are specified for data channels, and Polar codes, which are specified for control channels. The coding rate specifies the degree to which the redundancy is added, so lower coding rates imply the addition of greater amounts of redundancy.

D. Channel Models: AWGN and Rayleigh Fading

A realistic approach to analyzing system performance involves analyzing two channel models:

- **AWGN Channel:** Models white Gaussian noise as the primary impairment, providing a theoretical upper bound for performance.
- **Rayleigh Fading Channel:** Represents multipath fading environments with no dominant line-of-sight path, typical of urban mobile scenarios.

Such models allow us to evaluate the influence of fading and received noise on AMC performance under various channel conditions.

III. RELATED WORKS

Some research work has explored the topic of adaptive modulation and coding related to 5G NR communication.

Sharma et al. [6] presented a channel prediction scheme via deep learning, combined with AMC, designed for mMTC devices. Despite their proposed scheme being a learning-based approach, their results conclusively justified the significance of accurate channel prediction by experiencing a substantial improvement in BER compared to the traditional estimation via MMSE.

Mota et al. [7] proposed a reinforcement learning-based AMC scheme, QL-AMC, which learns an optimal strategy to select MCS to maximize spectral efficiency and decrease BLER. Although this work deals with traditional AMC, the reinforcement learning results are a point of comparison to establish the performance limit of the link.

A study by Chen et al. [8] explored a supervised learning (KNN)-based AMC strategy and confirmed that a better channel and noise estimation results in a better spectral efficiency and a lower BER. Even if a machine learning technique is not involved, the paper confirms the importance of a precise channel quality measurement in AMC again.

Contrary to the other approaches mentioned in the above list, this project maintains a standardized AMC operation as defined by 3GPP TS 38.214 [2], with MCS values regulating the Modulation and Code rates. The aim is to make a fair comparison between the link performance of 5G NR and how it can be simulated by MATLAB, assuming an AWGN and a Rayleigh channel.

IV. DESIGN AND EVALUATIONS

This section showcases the setup of the simulation as well as the evaluation framework we developed to analyze the **5G NR AMC**. We used EESM in order to compute the effective SNR, and we evaluate the performance under AWGN and frequency-selective Rayleigh fading. We replicate the link adaptation procedure defined in 3GPP TS 38.214, where we map as follows; *effective SNR* $\rightarrow$ *CQI* $\rightarrow$ *MCS*, to dynamically chose modulation order (Q_m) and coding rate (R) depending on the channel conditions.

A. Simulation Scenario:

In order to investigate the AMC performance, we employed and used a MATLAB environment. The code followed the procedures and conventions defined in the 3GPP TS 38.214 [2]. In addition, we use EESM model [9] to compute the effective SNR; we omit explicit interference, so in our setup SINR reduces to SNR. We also evaluate the AMC under two models: (i) **AWGN**, where all of the subcarriers have the same SNR, and (ii) **Rayleigh** fading, where all subcarriers will experience individual Rayleigh gains.

Within the simulation, we incorporate many fixed parameters, they are summarized as follows; the modulation schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM) as standardized from 3GPP TS 38.211 [4]. Furthermore, we use a bandwidth of **50 RB** at **30 kHz** SCS; a spatial layer count of **2**; and finally **0.5 ms** slot duration. The signal to noise ratio (SNR) is a swept variable taken from **-10 dB to 25 dB** in **0.5 dB** steps, and a coding rate range from **0.12** to **0.95** taken from the standardized MCS table. Each MCS value corresponds to a target coding rate R as well as a modulation order Q_m both defined within the 3GPP table mapping [2].

In order to model the SNR variability, we used two different cases. Within **AWGN**, all subcarriers have the same SNR, while in **Rayleigh** fading each subcarrier undergoes a different complex Gaussian gain. Per transmission, the subcarrier SNR are then combined using EESM to create a effective SNR using the formula:

$$\gamma_{\text{eff}} = -\beta \ln \left(\frac{1}{N} \sum_{n=1}^N e^{-\gamma_n/\beta} \right), \quad \beta = 1.0,$$

The effective SNR is mapped to a Channel Quality Indicator (CQI) using a threshold table derived from $\approx 10\%$ BLER points under AWGN [3]. Afterwards the appropriate MCS index is determined by the CQI via a mapping that is predefined in 3GPP TS 38.214.

In our simulation, we only focus on single-user downlink performance. For every SNR, we compute the per-subcarrier SNR (AWGN or Rayleigh), afterwards we compute the effective SNR via EESM, and map it into CQI, then from CQI to MCS where you obtain the Q_m and R . We chose to use a BLER model around the CQI threshold and compute throughput (T) directly:

$$T = \frac{n_L Q_m R N_{RE} (1 - \text{BLER})}{t_{\text{slot}} \cdot 10^6},$$

N_{RE} ($= 50 \times 12 \times 14 \times 0.8$) represents the amount of usable resource elements, however not all signals carry user data, so the overhead factor accounts for this (included in the formula, 0.8). The throughput is valued in terms of bits per slot where it is then normalized by slot duration (t_{slot}), which then yields bits per second (Mbps). The n_L scales throughput based on the number of special layers we have in the setup, which is 2 in our case, throughput grows proportionally to the layer count, so its vital that we include it in the setup formula.

B. Evaluation Settings:

The goal of the evaluation was to measure how AMC handles varying channel conditions within a controlled 5G NR downlink environment. We evaluate AWGN and Rayleigh fading and plot them for every metric. The only varying parameter in the study was average SNR, which we swept from -10 dB to 25 dB using a 0.5 dB step. Within each SNR value, we compute the effective SNR through EESM ($\beta=1.0$), we then determine the CQI (using approximately 10% BLER thresholds applied to the effective SNR), select the MCS based on that, and finally calculate the corresponding spectral efficiency and throughput. In addition, we incorporate a fixed MCS baseline to compare the AMC with (MCS 17: 64-QAM).

Throughput smoothing: In real world systems, short term throughput fluctuations is reduced via HARQ and outer-loop link adaptation (OLLA), these techniques are applied in deployed systems to adjust BLER variance and damp throughput variability. Instead of fully simulating HARQ/OLLA control loops, we emulate their effect by; (i) deriving the effective SNR through EESM which softens sub carrier variations; (ii) we also apply CQI/MCS hysteresis to avoid having the throughput ping-pong from high to low values between adjacent modulation levels; (iii) finally we perform light temporal smoothing on the throughput traces. This will yield a stable trend representative of a scheduler that applies HARQ and OLLA while keeping our simulation light.

To evaluate the performance of our implemented AMC algorithm, we define three different evaluations. Each evaluation focuses on a specific performance metric, which they

TABLE I: Simulation Parameters for AMC Evaluation

Parameter	Value	Description
Bandwidth	50 RBs	≈ 20 MHz NR channel
Subcarrier spacing	30 kHz	NR numerology
Layers	2	n_L in throughput
Slot duration	0.5 ms	t_{slot}
SNR range	-10 to 25 dB	0.5 dB step sweep
Channel models	AWGN, Rayleigh	Evaluation cases
CQI source	Threshold table	$\approx 10\%$ BLER ref. [3]
MCS selection	TS 38.214	CQI $\rightarrow$ MCS mapping
Modulations	QPSK-256QAM	TS 38.211 [4]
EESM β	1.0	SNR compression [9]
Fixed baseline	MCS 17 (64-QAM)	Reference curve

then combine together to collectively provide a comprehensive understanding of how the algorithm performs.

1) Evaluation 1 – CQI Selection as a Function of SNR:

Within this evaluation we investigate how CQI changes with SNR. By examining and reporting the CQI results, we are able to deduce whether our AMC mechanism can correctly interpret effective SNR values and assign CQI levels that align with standardized target BLER thresholds. This will reveal how accurately the system can capture current channel quality and transitions between link adaptation states.

2) Evaluation 2 – Modulation Order (Q_m) and Coding Rate (R): The modulation and coding configuration that is picked for each SNR value by AMC is evaluated here. By observing the changes in Q_m and R , we reveal whether the algorithm properly balances both reliability and throughput. This metric verifies the internal consistency of CQI-MCS mapping and adaptation logic.

3) Evaluation 3 – Downlink Throughput (Mbps): This evaluation focuses on the throughput per SNR level. Throughput in this case reflects the combined effect of modulation order, coding rate, and link reliability, which will provide a direct indication of how efficient AMC leverages channel capacity. Additionally, evaluating throughput also allows us to compare with a fixed MCS baseline to help quantify the gains from dynamic link adaptation.

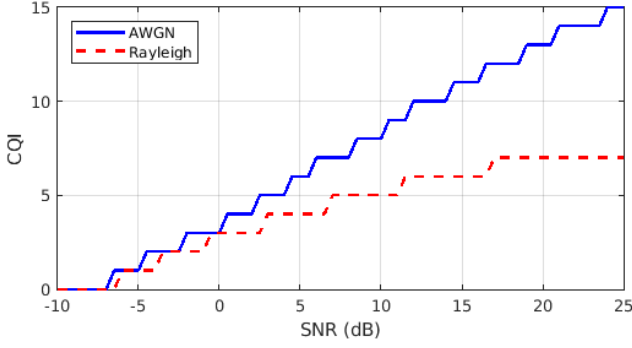
4) Overall: Combined, the three aforementioned evaluations will establish a complete framework for analyzing AMC in controlled 5G NR downlink conditions. It is important to note that our implementation didn't incorporate any channel mobility, path loss, or HARQ feedback. This ensures that our results will exclusively reflect the AMC behavior under this defined link conditions.

V. RESULTS

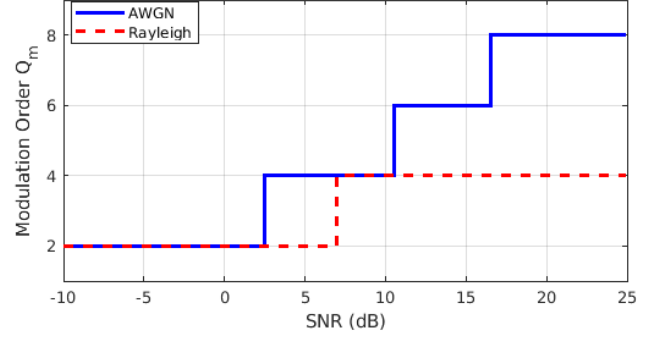
This section showcases the results of the three baseline evaluations discussed in Section IV-B. Each of the figures will correspond to one of the performance metrics previously defined in Section IV.

A. Evaluation 1 – CQI Selection as a Function of SNR

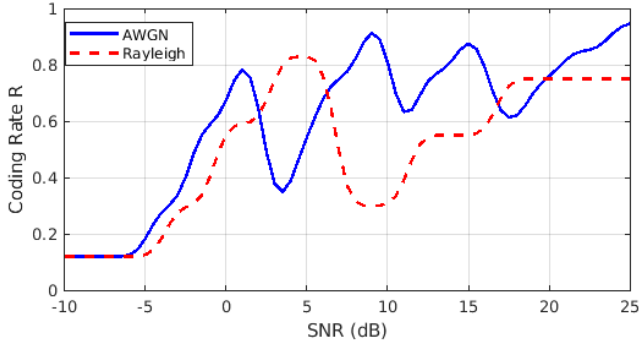
Figure 1a demonstrates the relationship between SNR and CQI. Unsurprisingly, the CQI index increases as SNR in-



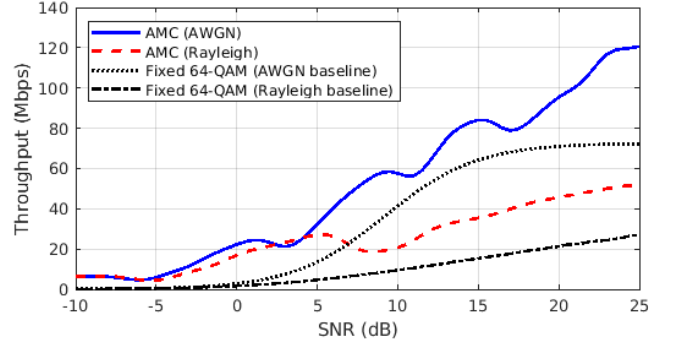
(a) Effective SNR $\rightarrow$ CQI mapping.



(b) Adaptive modulation (Q_m) selection.



(c) Adaptive coding rate (R) variation.



(d) Throughput comparison (Adaptive AMC vs. Fixed MCS).

Fig. 1: Evaluation results for AMC under AWGN and Rayleigh channel conditions.

creases, showing that effective SNR to CQI mapping correctly interprets channel quality within the implementation. The existing step-like progression in the graph represents the predefined CQI thresholds that were derived from $\approx 10\%$ BLER reference points, which confirms the link adaptation operates within the expected range. AWGN achieves larger CQI values for the same SNR, this is because AWGN is steady, while Rayleigh fades fluctuates across different sub-carriers, deep fades lower the averaged quality, as a result, the system chooses a lower and safer CQI for the same SNR. Consequently this will have an effect on when the modulation switches, which will be shown in the next evaluation.

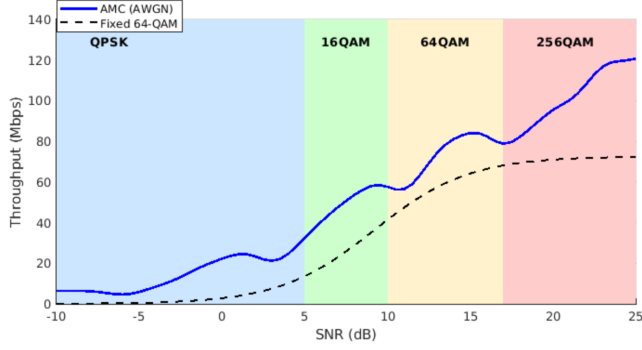
B. Evaluation 2 - Modulation Order (Q_m) & Coding Rate (R)

Next we present the adaptive modulation behaviour (Figure 1b). The modulation order (Q_m) for AWGN transitions in the following order as SNR improves (QPSK $\rightarrow$ 16QAM $\rightarrow$ 64QAM $\rightarrow$ 256QAM). This again confirms that our AMC implementation knows how to correctly select higher order modulations in the right channel conditions. Since Rayleigh achieves a lower CQI for the same SNR, there are only 2 modulation schemes that are used between -10 to 25 dB (QPSK $\rightarrow$ 16QAM). Figure 1c presents the coding rate as a function of SNR. For both AWGN and Rayleigh coding rate gradually increased with SNR, although it wasn't exactly linear due to discrete MCS steps, meaning that when the

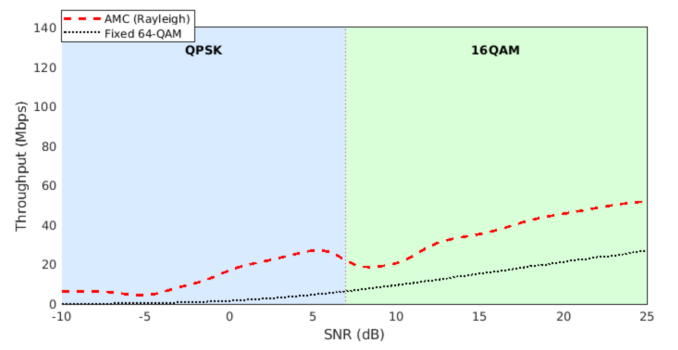
modulation switches the coding rate dips and then gradually increases again. Together, both of these figures verify that the algorithm closely follows the 3GPP defined MCS table.

C. Evaluation 3 – Downlink Throughput (Mbps)

Figure 1d showcases four different lines, AMC via AWGN, AMC via Rayleigh, and finally two fixed 64-QAM baselines (AWGN & Rayleigh) for same-conditions comparison. We plot two fixed 64QAM reference curves because one is for AWGN and the other is for Rayleigh, they are computing using the same EESM and BLER model as was used for the AMC curves. For each SNR, we average over 100 realizations; so for AWGN these realizations are pretty much identical, while for Rayleigh they provide a Monte Carlo averaging over independent fades, as a result this yields smoother results. At smaller SNRs, all four curves show small throughput values, however as the SNR improves, the adaptive scheme in AWGN quickly climbs much faster the rest. Where it beats Rayleigh and fixed 64-QAM (AWGN) from low to mid-SNRs and onwards, with this trend continuing at larger SNR values. Under Rayleigh fading, the curve grows slowly, nonetheless it generally stays below AWGN and above the fixed 64-QAM (Rayleigh) considering how the channel is less stable, so the system uses lower settings (modulation schemes) to play it safer. Both fixed 64-QAM lines performs poorly at low SNR, and then flattens to a constant throughput, this shows the



(a) AWGN: throughput vs SNR with modulation-region shading



(b) Rayleigh: throughput vs SNR with modulation-region shading

Fig. 2: Throughput comparison under AWGN and Rayleigh channels.

disadvantages of using 64-QAM as it wastes headroom at both high and low SNRs. The fixed modulation line also shows no fluctuations because it never switches mode; with no threshold crossings the rate stays flat once SNR is adequate. The small zig zags on the AWGN and Rayleigh graph is expected, as each time the system steps up to a higher mode it briefly carries more errors, so the net rate temporarily dips and soon after recovers as SNR keeps increasing. Overall, AMC ensures a higher average throughput compared to a fixed modulation scheme.

D. Overall

Figure 2 showcases when the modulation switches occur and provides perspective on why there are dips at certain intervals. When the modulation scheme changes on both graphs there is a slight temporary dip in performance, and this is because when you switch to a larger modulation scheme you briefly carry more errors so throughput drops, this soon solves itself and throughput quickly increases soonafter.

VI. CONCLUSIONS

The study evaluated a Adaptive Modulation and Coding implementation in 5G NR using EESM model. By following the standard MCS and CQI definitions found in the 3GPP TS 38.214 we can ensure that our simulation accurately represents link adaptations under AWGN and Rayleigh conditions.

There was a consistent relationship between the SNR and CQI, modulation order, coding rate, and throughput. The AMC successfully adjusted the parameters of transmission to ensure link reliability and optimized spectral efficiency. The fixed MCS configuration had a lower throughput all around while the adaptive scheme achieved a much higher throughput on average for both AWGN and Rayleigh.

These findings demonstrate a valid implementation of AMC that adheres to the defined standards while showcasing the performance results of AMC compared to a fixed MCS. Future work includes adding mobility and path loss to map SNRs to distance, additionally it could be interesting and beneficial to add HARQ in the simulation to evaluate AMC under more realistic link adaptations.

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1) MCS and CQI Definition:

```

1  %% =====
2  % 5G NR Adaptive Modulation & Coding (Ultra-Smooth Version
3  )
4  % Includes EESM, CQI hysteresis, and fixed MCS comparison
5  % =====
6  clear; clc; close all;
7
8  %% 1. MCS Table and CQI Mapping
9  mcsTable = [
10     0 2 0.12; 1 2 0.19; 2 2 0.30; 3 2 0.44; 4 2 0.59; 5 2
11     0.74;
12     6 2 0.83; 7 2 0.90; 8 2 0.95; 9 4 0.30; 10 4 0.43; 11 4
13     0.55;
14     12 4 0.66; 13 4 0.75; 14 4 0.83; 15 4 0.90; 16 4 0.95; 17
15     6 0.45;
16     18 6 0.55; 19 6 0.65; 20 6 0.77; 21 6 0.84; 22 6 0.90; 23
17     6 0.95;
18     24 8 0.60; 25 8 0.75; 26 8 0.85; 27 8 0.95];
19  Qm_list = mcsTable(:,2);
20  R_list = mcsTable(:,3);
21
22  cqiThresh_dB = [-Inf -6.7 -4.7 -2.3 0.2 2.4 4.3 5.9 8.1
23     10.3 11.7 14.1 16.3 18.7 21.0 23.6];
24  cqi2mcs = [0 0 2 4 6 9 11 13 16 18 20 22 24 25 26 27];
25
26  %% 2. System Parameters
27  SNR_dB = -10:0.5:25; nRB = 50; nLayers = 2;
28  slotDur_s = 0.5e-3; overhead = 0.8;
29  nRE = nRB*12*14*overhead; Nsub = nRB*12;
30  beta = 1.0; nReal = 100; a = 0.35;
31  channels = {'AWGN', 'Rayleigh'};
32
33  %% 3. Adaptive Simulation (CQI, BLER, Throughput)
34  for c = 1:2
35     channelType = channels{c};
36     prevCQI = 0;
37     CQI = zeros(size(SNR_dB)); Qm = CQI; R = CQI; Thru_Mbps
38     = CQI;
39
40     for i = 1:numel(SNR_dB)

```

Listing 1: MCS Table and CQI Thresholds for 5G NR AMC

2) System Parameters and Channel Setup:

```

41     switch channelType
42     case 'AWGN'
43         gamma_dB = SNR_dB(i) * ones(1,Nsub);
44     case 'Rayleigh'
45         ray = (randn(1,Nsub)+1i*randn(1,Nsub))/
46             sqrt(2);
47         gamma_dB = SNR_dB(i) + 20*log10(abs(ray
48             ));
49     end
50     gamma_lin = 10.^(gamma_dB/10);
51     gammaEffLin = -beta * log(mean(exp(-gamma_lin/
52         beta)));
53     gammaEff_dB_all(r) = 10*log10(gammaEffLin);
54 end
55 gammaEff_dB = mean(gammaEff_dB_all);
56
57 % CQI update with hysteresis
58 k = find(gammaEff_dB >= cqiThresh_dB, 1, 'last');

```

Listing 2: System Parameters and Channel Initialization

```

59     if abs(cqiNew - prevCQI) > 0.5
60         CQI(i) = cqiNew; prevCQI = cqiNew;
61     else
62         CQI(i) = prevCQI;
63     end
64     CQI(i) = min(CQI(i), 15);
65
66     % Map CQI -> MCS and compute throughput
67     mcsIdx = cqi2mcs(CQI(i)+1);
68     Qm(i) = Qm_list(mcsIdx+1);
69     R(i) = R_list(mcsIdx+1);
70     thr = cqiThresh_dB(CQI(i)+1);
71     BLER = 1 ./ (1 + exp(a*(gammaEff_dB - thr)));
72     SE = Qm(i)*R(i);
73     bitsPerSlot = nLayers * SE * nRE;
74     Thru_Mbps(i) = bitsPerSlot * (1 - BLER) / slotDur_s
75         / 1e6;
76 end
77
78 % Smoothed results for visualization
79 R = smoothdata(R,'gaussian',7);
80 Thru_Mbps = smoothdata(Thru_Mbps,'gaussian',9);
81 results(c).CQI = CQI; results(c).Qm = Qm;
82 results(c).R = R; results(c).Throughput = Thru_Mbps;
83 end
84
85 %% 4. Fixed 64-QAM Baseline (MCS17)
86 fixedMCS = 17;
87 QmF = Qm_list(fixedMCS+1); RF = R_list(fixedMCS+1);
88 SEF = QmF*RF;
89 bitsPerSlotF = nLayers * SEF * nRE;
90 thr_mcs17 = 0.5*(cqiThresh_dB(9)+cqiThresh_dB(10)); % 9.2
91 dB
92
93 Throughput_fixed_AWGN = zeros(size(SNR_dB));
94 Throughput_fixed_Rayleigh = zeros(size(SNR_dB));
95
96 for c = 1:2
97     channelType = channels{c};
98     for i = 1:numel(SNR_dB)
99         for r = 1:nReal
100             switch channelType
101             case 'AWGN'
102                 gamma_dB = SNR_dB(i)*ones(1,Nsub);
103             case 'Rayleigh'
104                 ray = (randn(1,Nsub)+1i*randn(1,Nsub))/
105                     sqrt(2);
106                 gamma_dB = SNR_dB(i) + 20*log10(abs(ray
107                     ));
108             end
109             gamma_lin = 10.^(gamma_dB/10);
110             gammaEffLin = -beta * log(mean(exp(-gamma_lin/
111                 beta)));
112             gammaEff_dB_all(r) = 10*log10(gammaEffLin);
113         end
114     end
115 end

```

Listing 3: EESM Averaging, CQI Hysteresis, and Throughput Computation

4) Fixed MCS Reference Calculation:

```

116     gammaEff_dB = mean(gammaEff_dB_all);
117     BLER_fixed = 1 ./ (1 + exp(a*(gammaEff_dB -
118         thr_mcs17)));
119     Thru_fixed = bitsPerSlotF * (1 - BLER_fixed) /
120         slotDur_s / 1e6;
121     if c == 1
122         Throughput_fixed_AWGN(i) = Thru_fixed;
123     else
124         Throughput_fixed_Rayleigh(i) = Thru_fixed;
125     end
126 end
127 end

```

Listing 4: Fixed 64-QAM (MCS 17) Baseline Throughput Calculation